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Detection of Driver Drowsiness Using Adaptive Eye Characteristic Ratio for Enhanced Road Safety

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ABSTRACT Every year, countless people lose their lives in serious car accidents, and drowsy driving is a major cause. However, because the earliest indications of exhaustion can be identified before a dangerous scenario develops, the detection and signaling of driver drowsiness are ongoing research topics. This study presents a real-time detection system designed to detect driver's sleep and deliver timely warnings. The mechanism employs an eye monitoring system that uses eye feature points to detect whether a driver's eye is tired and delivers an alert if the driver is drowsy. The proposed work recognizes the driver's face by collecting facial landmarks and gives warning to the driver to avoid crashes by performing adaptive thresholding and drowsiness diagnoses based on Eye Characteristic Ratio. The empirical results show that using random forest classifier, decision tree classifier, KNN classifier, and multinomial classifier, the proposed model achieved 90.48%, 89.44%, 88.63%, and 84.32% accuracy, respectively.

INDEX TERMS Drowsiness, Adaptive Eye Characteristic Ratio, Landmarks, Eye Blink

I. INTRODUCTION

Driving long distances, especially on highways, can be exhausting and tiresome. Drowsiness have been the leading causes of serious traffic accidents in India and throughout the world in recent years. The rise in the percentage of traffic accidents caused by drowsiness and exhaustion caught the attention of the researchers. It has also been shown that when tiredness increases, drivers' performance deteriorates. The number of deaths and injuries in traffic accidents caused by human error is increasing every year. Drowsiness and driving are very dangerous and difficult to recognize [1]. Sleep deprivation or a tired sensation can reduce attention and focus, making it difficult for drivers to identify impending danger. When a driver is not paying attention while driving, their response time slows down, and their steering behavior deteriorates. Driver drowsiness will cause various physical and economic losses as the decision-making process and reaction speeds are hindered. All these factors can contribute to exacerbating accidents. Driver drowsiness detection is an automobile safety feature that aids in the prevention of accidents caused by sleepy driving. Fatigue driving will significantly increase the chance of collision. During dangerous times, it will poorly affect the driver's ability to react safely and quickly, while drowsiness will also reduce the driver's activity and his understanding of what's happening on the road decreases and driver may lose attention at this point and be unable to take steps such as avoiding a frontal collision or halting on time. If a car travels at 100 km/h and the driver falls asleep

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for around 4 seconds, the automobile will go for more than 100 meters, i.e., the car will travel for more than 100 meters without the driver's control. As per the National Sleep Foundation (NSF) data, 51% of young people feel sleepy while driving, and 17% fall asleep [2] and 40% of highway accidents occur due to drivers dozing off. According to the reports based on extensive accident investigations, fatigue is implicated in 10 to 20% of major traffic accidents [3]. Based on a study, the automobile industry, academic institutes, and other governmental organizations are developing solutions to prevent such catastrophes, given that approximately 1.4 million people die on the road every year, making it the 7th leading cause of death in 2016 [4]. As signals generated by daily physical activities can be used to determine a human's physiological or psychological state, most existing drowsiness detection algorithms/systems are based on bio-electrical signal processing or image/video processing to measure the driver's physiological and behavioral information [5]. Drowsiness and distraction are factors in a considerable proportion of car accidents. According to the AAA Foundation for Traffic Safety's study brief, tiredness was detected in 10.6 per cent –10.8 per cent of collisions that resulted in minor property damage, airbag deployment, or injury in real-world driving tests [6]. To reduce casualties and damage, installing a driver drowsiness detecting system within the vehicle is critical. This section discusses various methods of measuring the driver's drowsiness which comprise vehicle-based measurements, behavioral measurements, and physiological measurements [7]. All these methods are explained below.

A. Subjective Measurement

In this method, drowsiness is measured based on the driver's estimate. Various techniques have been used to rate driver drowsiness under subjective measurement—for Example, Karolinska Sleepiness Scale, Epworth sleepiness scale.

B. Vehicle-based measurement

Vehicle-based measurement is another approach to measure driver drowsiness. In this approach, sensors are fitted at various locations in the vehicle, such as the steering wheel and accelerator pedal; drowsiness is detected as per the signal sent by the sensors. For Example, Steering Wheel Movement (SWM).

C. Behavior measurement

The behavior of the drowsy driver was observed as head posture, frequent yawning, closing eyes, blinking, etc. The camera detects whether the driver is drowsy vehicle-based and if any of these conditions occur, the driver will be alerted.

D. Physiological measures

Physiological measures are not commonly used. Physiological signals are the most accurate in detecting a drowsy driver. It is a direct measure that can stop many accidents caused by driver drowsiness. Electrocardiogram, electromyography, electroencephalogram and electrooculogram, heart rate are possible measurements of physiological signals. This project focuses on ambient computing which is a broad term that describes a collection of hardware, software A.I. decisions, and human activity that enables computer actions without the intervention of direct human commands.

Section II provides a quick overview of the existing related study and work. The proposed method, which incorporates face detection, facial landmarks detection, eye detection, eye blink detection, adaptive thresholding, and drowsiness detection, is described in Section III. In Section IV, the experimental findings on the dataset are reported. Finally, Section V summarizes the outcomes and results and brings the work to an end.

II. Related Study

With the rapid advancement of technology, it is now feasible to create an accurate system by employing various methods to identify the early stages of fatigue in drivers and reduce the chance of an accident. This section provides a study of some of the already existing systems [26] to detect drowsiness to help prevent accidents. Computer vision is being used to recognize human actions in the video (CV) [8]. As a result, image/video processing may monitor operator performance to guarantee proper job execution and extract fatigue-related features in real-time [9]. The system captures videos and acknowledges the driver's face in every frame using image processing techniques. Face alignment, head posture estimation, and blink detection have been effectively accomplished using facial landmarks [10]. Facial landmarks are used to locate and track various facial features such as the eyes, eyebrows, nose, mouth, and jawline. Face landmark detection is a subset of the shape prediction. A shape predictor tries to find significant areas of interest along with a shape (and typically an ROI that describes the item of interest) [11]. Dlib is a popular and widely used library for extracting face landmarks. This library's face detector is based on histograms of oriented gradients. The 68(x, y)-coordinates corresponding to facial structures on the face are estimated using the pre-trained facial landmark detector inside the dlib package. The 68-point iBUG 300-W dataset was used to train the dlib facial landmark predictor. The pre-trained face detector in Dlib is based on a variant of the traditional Histogram of Oriented Gradients + Linear SVM object identification method. The system recognized facial landmarks and computed the Eye Aspect Ratio (EAR) and Eye Closure Ratio (ECR) to identify driver fatigue. To assess the efficacy of the suggested technique, machine learning techniques were used. Driver's tiredness can be identified using a device that works on image processing using a camera that focuses on the driver to prevent traffic accidents [12]. The changes in the face can be assessed and then run through a program to

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identify tiredness and provide an alarm to the driver. Calculating modified EAR detects blinking eyelids and indicates whether the driver is drowsy [13]. According to research [14], for the condition of drowsiness, if the number of blinks per minute is less than 10, the individual is drowsy. The blink pattern or blink rate is critical for monitoring a human operator's attentiveness [15]. The blinking frequency of a person's eyes varies when they are fatigued. The blink rate may be reduced to 4-6 blinks per minute compared with a sleep-deprived person's blink rate [16]. A system that detects eyes and calculates the eye-blink rate using the identified eyes, and if it lowers from the usual threshold level, it calls the person tired. Viola-jones cascade classification method for eye detection and MATLAB for implementing the technique has been employed [17]. Various computer vision applications have been constructed to calculate blink rate, eye closure, and yawning to detect a driver's tiredness while driving successfully and swiftly [18]. The Smart Vehicle System (SVS) was built to identify a driver's drowsiness and weariness in real-time based on the image obtained. The study was based on behavior analysis, high-end camera installation, and a traditional method to discover probable coordinates for identifying eyes and mouth. Working with NTHU-DDD Dataset, an adaptive descriptor with the virtues of uniqueness and robustness is generated from an updated version of HOG features derived from binarized histograms of shifting orientations which is offered as a new framework for driver drowsiness detection [19]. It also mentions that light variation was eliminated by using infra-red (IR) cameras. The final HOG descriptor produced from binarized HOG features was passed to the trained Naive Bayes (NB) classifier to obtain the final driver drowsiness conclusion. The data based on the image obtain drowsiness detection accuracy also provided an experimental performance comparison of the framework versus several state-of-the-art techniques. Significant artificial intelligence-based algorithms were employed to identify driver tiredness and the pace at which the driver is sleepy [20]. To identify driver tiredness, an algorithm was presented that analyses vertical distances between the eyes and mouth, eye closure, yawning, and other artificial facial traits. The performance of several machine classifiers, for example K-nearest neighbor, support vector machine, and artificial neural network classifiers for detecting driver tiredness with varied road geometries (straight segments and curve segments) has been compared [21]. A multilayer ensemble metric is used to characterize drowsiness, substantially and financially cheap input source, and state-of-the-art classification approaches [22]. The DDD decisions were made by extracting meaningful features from the driver's eye. Features extracted are accumulated over a stipulated period to generate measurable variables for the actual drowsiness detection. These include a face and eye state detection model and the calculation of the percentage of eyelid closure (PERCLOS), blink frequency (BF), and maximum closure duration (MCD) as the selected features. These features were then fed as inputs into several machine learning models for drowsiness classification. Models from the K-nearest Neighbors (KNN), Support Vector Machine (SVM), Logistic Regression, and Artificial Neural Networks (ANN) machine learning algorithms were experimented [23]. Using CNN networks, a semi-VGG architecture is used to anticipate whether the eyes are open or closed. The decision to notify the driver is made by eye closure detection methods. It is a system that employs RGB-D cameras to detect drowsiness and, as a result, offer warnings based on the circumstances met. Two significant modules were employed concurrently to evaluate the condition of the driver's mouth and eyes to identify drowsiness. Depth information is used to determine the mouth area and its condition at first [24]. The circular Hough Transform allowed the identification of the iris center and points of intersection for the two lids to operate the two geometric descriptors. Following the extraction of two geometric descriptors, the signal was non-linear and non-stationary. The SVM approach validates the study to achieve the results. Instead of utilizing video-based techniques for driver drowsiness detection, a method has been devised to compute the ECD using EEG sensors [25]. The idea was to bring changes in the occipital EEG to have a linear connection with the ECD. The ECD estimate technique made ECD estimation more efficient. A related comparative survey is portrayed in table 1.

The deep learning approach has garnered increasing popularity owing to its capability to automate the feature selection and extraction processes, which would otherwise necessitate manual intervention [37]– [40]. Additionally, the models utilized in deep learning not only extract features but also possess the ability to learn from those features with distinct qualities, allowing for various applications, including classification, detection, and segmentation. Two prominent examples of architectures employed in deep learning are convolutional neural networks (CNN) and deep neural networks (DNN). CNNs are frequently trained for differentiating between images [28], [41]– [44] and are composed of convolutional layers that facilitate feature extraction from images, culminating in fully connected layers. Consequently, these models are effective in extracting distinguishing features among labeled images. Conversely, DNNs are characterized by multiple interconnected neurons, with each neuron linked to every neuron in both the preceding and subsequent layers, and a "weight" value that influences the neuron's contribution to the overall architecture [45], [46]. These weights undergo updates throughout the training phase.

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TABLE 1
RELATED WORK SUMMARY

Year of Publication	Author	Methodology and Dataset	Parameters
2019	Drutarovsky and Fogelton [29]	Blinks are recognized by examining the alteration of the vertical motions in the eye region(Dataset: ZJU)	93.45% accuracy
2016	Zeeshan Ali Haq Ziaul Hasan [18]	Eye-Blink rate detection (BioID dataset)	87% accuracy
2021	Samy Bakheet, Ayoub Al-Hamdi [20]	Improved HOG Features and Naïve Bayesian Classification (Dataset: NTHU-DDD)	85.62% accuracy
2019	Ameen Aliu Bamidele , Kamilia Kamardin , Nur Syazarin Natasha Abd Aziz , Suriani Mohd Sam Irfanuddin Shafi Ahmed , Azizul Azizan , Nurul Aini Bani , Hazilah Mad Kaidi [34]	Face and Eye Tracking (Dataset: National Tsing Hua University (NTHU))	72.25% accuracy
2019	Mina Zohoorian Yazdi, Mohsen Soryani [25]	Yawning and Eye Closure (VAP_RGBD dataset)	89% accuracy
2018	Lamba and Virmani [16]	Eye blinks are detected using a 13 dimensional temporal window of the eye facet correlation (Dataset: ZJU)	97.6% accuracy
2020	Kepesiova et al. [35]	A convolutional neural network, a convolutional control gate-based recurrent neural network, and a voting layer were used (annotated dataset)	Average accuracy 84.41%
2019	You et al. [36]	Deep cascaded convolutional neural network (Dataset: FDDB)	Average accuracy 94.80%
2023	Safarov, F [39]	eye-gaze landmarks to measure blinking.	Average accuracy 85%

III. Materials and Methods

The following is a combination of two-stage work for detecting driver drowsiness. The first section uses the Dlib package to identify face landmarks, and a threshold value is calculated to assess whether the driver is drowsy. Following that, employing the 68 facial landmarks, the ECR (Eye Characteristic Ratio) is determined. Also, the number of eyes blinks every 60 seconds is calculated. If the ECR value falls below the threshold, it means the person is sleepy. In the event of drowsiness, an alarm would sound, alerting the driver and passengers. In the next section, four classifiers, namely Random Forest, Decision tree, KNN, and Multinomial Classifiers, are applied to determine the accuracy of all the classifiers and present the top classifier for our system amongst the four.

Section III(A) provides detailed information about the chosen dataset. There are two subsections in this section. Subsections provides information on data processing, i.e., the entire work scenario, and provides a thorough perspective of the data segregation technique utilized in our study. Section III(B) illustrates the performance parameters that will be assessed to determine the efficiency and accuracy of the created model. The last part, Section III(C), provides a detailed description of the whole experimental setup for the model's operation.

A. Dataset

The only publicly accessible video dataset offering frame-level annotations for tiredness, head, eyes, and mouth condition, NTHU Drowsy Driver Detection (NTHU-DDD) [25], was utilized to assess the proposed Multimodal Driver drowsiness detection system. National Tsing Hua University generated the NTHU-DDD dataset (table 2), utilized at the Asian Conference on Computer Vision Workshop on Driver Drowsiness Detection. This video dataset features 36 persons of various ages, genders, and ethnicities in 5 scenarios with various physical characteristics like skin tone, fatigue, facial structure, clothing, and hairstyles. As a result, the algorithm for sleepiness detection should consider robustness and efficiency under all conditions. A training dataset, an evaluation dataset, and a testing dataset make up the entire dataset. Though table 2 defines three labels, only two labels (normal and drowsy) are used in the work and the label defining the scenario where eyes are looking aside are neglected.

The training dataset includes 360 video clips (722,223 frames) of 18 people, 10 males and 8 females. The testing dataset contains 70 video clips (736,132 frames) of 14

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people, whereas the assessment dataset has 20 video clips (173,259 frames) of four subjects. Each frame is labelled sleepy or non-drowsy throughout training and evaluation. Glass, no glasses, glasses at night, no glasses at night, and sunglasses are the five situations in which each volunteer serves as a driver. The volunteer records four videos with distinct content in each scene: non-sleepy, sleepy states combining three usual states, blinking with nodding and yawning, and non-sleepy, tired states combining three typical states, blinking with nodding and yawning. Each video features four annotation files, one for sleepiness, one for eyes, one for the head, and one for the mouth. The video data for the night (glasses) and night (no glasses) sequences were filmed at 15 frames per second, while the video data for the other scenes were shot at 30 frames per second. All video frames are grayscale pictures with a resolution of 640*480 pixels and are devoid of audio data. We only utilized the training and assessment datasets in this research because the testing dataset's fundamental authenticity label has not been revealed. Figure 1 shows different snapshots from the NTHU drowsy driver detection (NTHU-DDD) dataset having different facial expressions. The first person is smiling and is not drowsy. The second person's eyes are closed thus he is drowsy. The third person is yawing thus tired, and the fourth person's head is nodding.



FIGURE 1. Dataset

Here input videos are classified into two categories-

- Training videos, which are used for the training of classifiers.
- Testing videos, which are used to verify the algorithm by predicting the drowsiness of the driver.

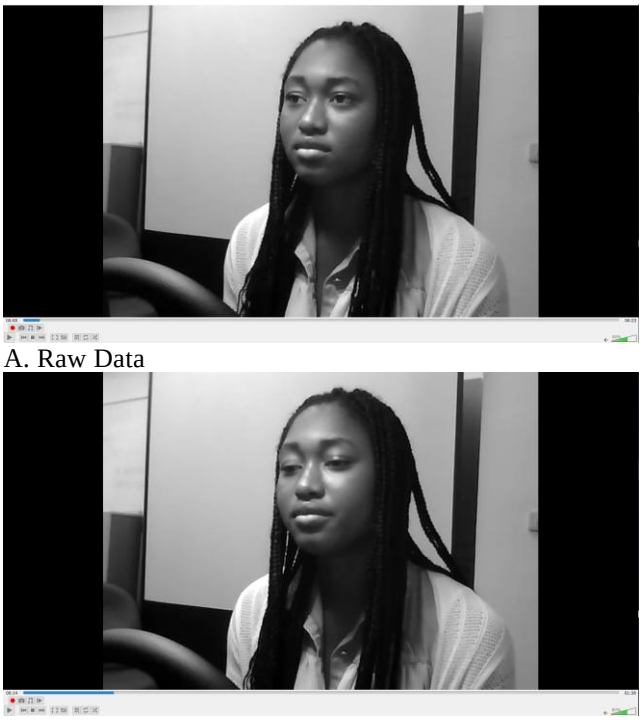
TABLE 2
NTHU-DDD DATASET SCENARIOS

Label	Sub- Category			
	Drowsiness	Eyes	Head	Mouth
0	Normal	Normal	Normal	Normal
1	Drowsy	Sleepy	Nodding	Yawning
2			Looking aside	Talking and laughing

1) DATASET PRE-PROCESSING

The videos of the NTHU-DDD Dataset are fed to the system. From the videos, image frames are obtained. Figure 2a represents the snapshot of the actual video with a duration of 1 minute and 30 seconds at a particular instance. Figure 2b is another snap of the same video, which is cropped for an initial 30 seconds after data pre-processing.

For the requirement of our work, we took the first 30 seconds of the complete video to perform adaptive thresholding on the subject. In the event of the real-time implementation scenario, our strategy for our specific model is as follows:



A. Raw Data

B: Processed Data

FIGURE 2. Figure 2a represent the raw data and figure 2b show the processed data

The driver is instructed to close his eyes for 15 seconds to obtain his lowest eye characteristic ratio and then open his eyes for another 15 seconds to determine his highest eye characteristic ratio. The average ratio is calculated by taking the mean of the 50 highest and 50 lowest ratios and is set as the threshold value for the driver. This threshold will be stored in the system, and the system will identify tiredness based on blinks and drowsiness based on ECR throughout the drive. We attempted to apply the real-life approach to the dataset video. The precise frames, i.e., 900 frames (in the pattern of 0 and 1) corresponding to those 30 seconds, are calculated. After the adaptive thresholding is done and the threshold is calculated, the complete video of the subject is tested. Based on that value, the driver is monitored during the entire video.

2) DATA SEGREGATION

Since the dataset contains numerous characteristics, such as nodding, yawing, and several more, we extracted the blinking and eye opening-closing section from the dataset videos to calculate threshold value and correspondingly alternated the text files connected to it. We examined all the videos and

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selected the most appropriate subjects for our work in the ratio of 70:30 for training and testing, respectively. Descriptions of some of the testing subjects are mentioned in Table 3.

TABLE 3
NTHU-DDD DATASET (SEGREGATED)

Subject	Gender	Beard or moustache	Glasses
1	Male	No	No
2	Female	No	No
3	Male	Yes	No
4	Male	No	No
5	Male	No	No
6	Female	No	No
7	Female	No	No
8	Female	No	No
9	Male	No	Yes
10	Female	No	Yes

B. Performance Parameters

In this work, drowsiness detection is accomplished using classification. The performance of the classification algorithms employed are compared using many parameters (where TP stands for True Positive, TN stands for True Negative, FP stands for False Positive, and FN stands for False Negative), as shown below:

- *Sensitivity* is defined as the percentage of accurately classified examples of a class to the overall number of occurrences in that class

$$SE = \frac{TP}{TP + FN} \quad (1)$$

- *Specificity* is a metric that counts the percentage of true negatives accurately identified as such.

$$SP = \frac{TN}{FP + TN} \quad (2)$$

- *Precision* is defined as the proportion of correct positive predictions to total positive predictions

$$P = \frac{TP}{TP + FP} \quad (3)$$

- *Negative Predictive Value (NPV)* is the fraction of true negative findings to total negative predictions in statistics and diagnostic testing.

$$NPV = \frac{TN}{TN + FN} \quad (4)$$

- *False Positive Rate (FPR)* is the ratio of erroneous positive predictions to the total number of negatives

$$FPR = \frac{FP}{FP + TN} \quad (5)$$

- *False Discovery Rate (FDR)* is the proportion of null hypothesis rejections that turn out to be false rejections

$$FDR = \frac{FP}{FP + TP} \quad (6)$$

- *False Negative Rate (FNR)* is the proportion of erroneous negatives to total positives predicted.

$$FNR = \frac{FN}{FN + TP} \quad (7)$$

- *Accuracy (ACC)* is defined as the proportion of correct predictions to total occurrences.

$$ACC = \frac{TP+TN}{P + N} \quad (8)$$

- *F1 Score (F1)* is a combination of accuracy and recall that's been weighted.

$$F1 = \frac{2*TP}{(2*TP + FP + FN)} \quad (9)$$

- *Recall* quantifies the number of positive class predictions made from all positive examples in the dataset.

$$RECALL = \frac{TP}{TP + FN} \quad (10)$$

- *Mean Squared Error (MSE)* is calculated as the average of the squared differences between the predicted and actual values.

$$MSE = \frac{\sum_{i=1}^{N}((Predicted_i - Actual_i)^2)}{N} \quad (11)$$

- *Root Mean Squared Error (RMSE)* is a standard method of calculating a model's error in predicting quantitative data.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}((Predicted_i - Actual_i)^2)}{N}} \quad (12)$$

- *Matthews Correlation Coefficient (MCC):* The correlation coefficient generated from the four CM values is known as the Matthews Correlation Coefficient (MCC) (confusion matrix).

$$MCC = TP * TN - \frac{FP * FN}{\sqrt{(TP + FP) * (TP + FN) * (TN + FP) * (TN + FN)}} \quad (13)$$

- *Receiver operating characteristics (ROC)* analysis gives methods for identifying the best models and eliminating the ones that aren't.

- *Area under curve (AUC)* is the area under the ROC Curve.

C. Experimental Setup

A system is presented to detect the driver's drowsiness that works on grayscale videos. The scheme of the system is shown in Figure 3. The initial stage in the stepwise depiction of the processes for the model is to acquire a video frame. After acquiring the video frame, the following step is to detect the driver's face. If a face is detected, the system will follow and monitor the eyes to determine the adaptive threshold value for the eye pair. It will then constantly scan eyes frame by frame to compute ECR, which is then passed to the set condition of ECR < threshold. If the condition is satisfied, it diverges into two parts. The first section checks if the result was achieved for less than 60 or more than 60 frames, which causes the algorithm to move to the next step while if the condition is not met, the algorithm resumes detecting the driver's face. When the blink count is less than 10 in 60 seconds, a tiredness

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warning appears, and a drowsiness alert appears when ECR < threshold for 60+ frames.

1) STEP 1- VIDEO TAKEN FROM THE DATASET

A video is chosen from the dataset and is fed to the model for further processing. Figure 4 depicts the output of the video taken for the entire working of the model.

2) STEP 2- FRAME EXTRACTION FROM THE VIDEO

Using OpenCV methods, frames are retrieved from the video captured in Step 1. Figure 5 shows a depiction of the video frames.

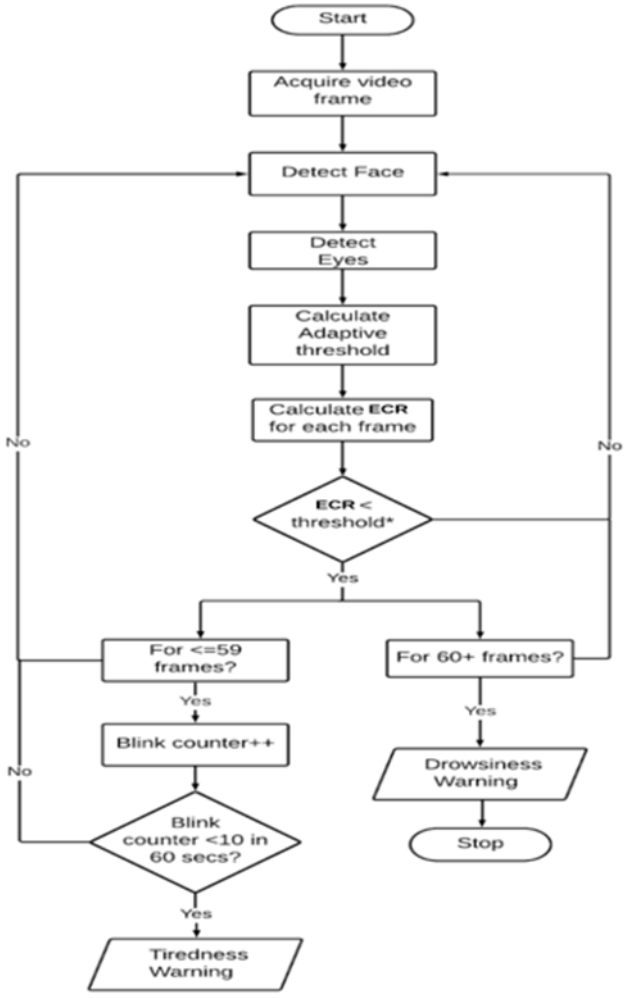


FIGURE 3. Experimental Setup

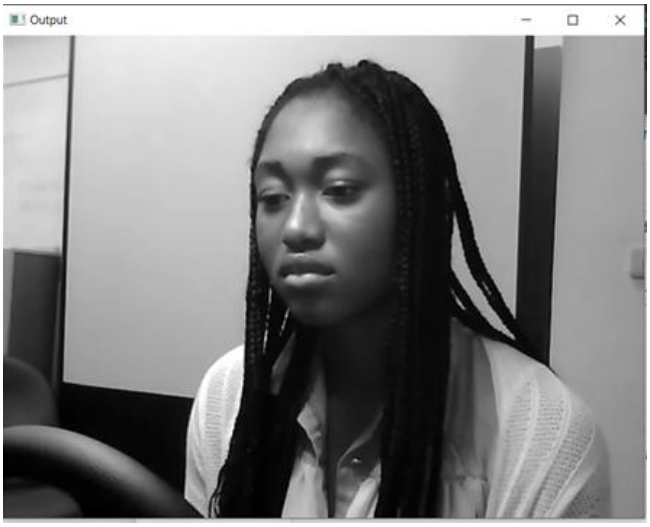


FIGURE 4. Video Detection



FIGURE 5. Frame Extraction

3) STEP 3- DETECTION OF THE FACE USING DLIB LIBRARY

The method begins by extracting the facial area from the input video frame. For this purpose, dlib.get_frontal_face_detector () has been used. It creates a front face detector based on the histogram of directional gradient characteristics (HOG) and linear classifier. In particular, the HOG trainer uses a structural training algorithm based on SVM, which allows the trainer to train in all sub-windows of each training image. The dlib.get_frontal_face_detector () function does not accept any parameters. It returns the pre-trained HOG + Linear SVM face detector from the dlib library when called [27]. The face region was successfully recognized in the video by using dlib.get_frontal_face_detector(). Figure 6 depicts the end outcome.

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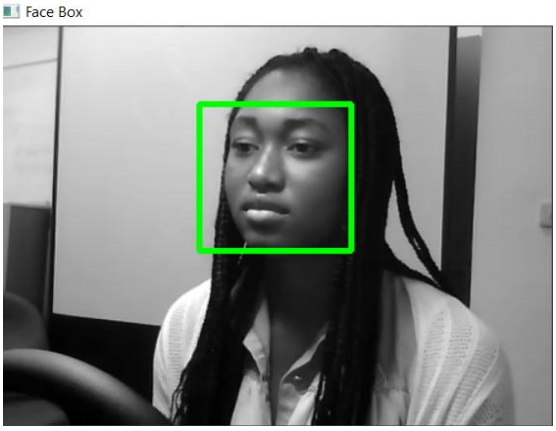


FIGURE 6. **Face Detection**

4) STEP 4- PLOTTING OF 68 LANDMARKS ON THE FACE

Once the face has been recognized, the next step is to locate the 68 landmarks using a shape predictor, as shown in Figure 7 [11]. Before that, the facial pictures should be normalized to reduce the effect of camera distance, non-uniform lighting, and variable image resolution. As a result, the facial image is resized to 500 pixels in width and converted to grayscale. The Dlib package includes a facial landmark detector based on the paper by Kazemi and Sullivan's One Millisecond Face Alignment using an Ensemble of Regression Trees. The dlib pre-trained model attempts to locate and identify the following face areas, yielding 68-point coordinates to estimate their position:



FIGURE 7. **Shape predictor 68 facial landmarks**

- The jaw is accessed using coordinates [0, 16].
- The right eyebrow is accessed using coordinates [17, 21].
- The left eyebrow is accessed using coordinates [22, 26].
- The nose is accessed using coordinates [27, 35].
- The right eye is accessed using coordinates [36, 41].
- The left eye is accessed using coordinates [42, 47].
- The mouth is accessed using coordinates [48, 67].

The result is utilised for high-quality facial landmark detection. Figure 8 shows the position of 68 (x, y) landmark coordinates returned by the pre-trained shape predictor. The next step is to segregate the landmarks of the eyes and measure the distances between them for further process.

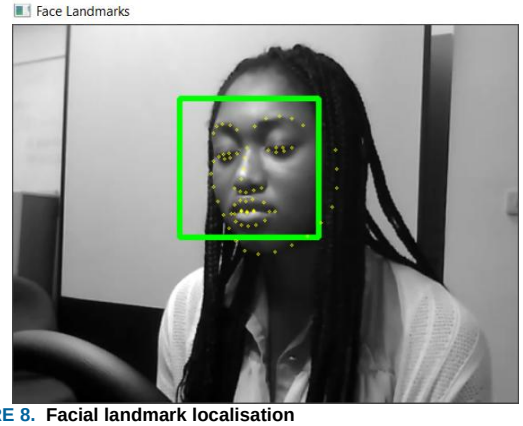


FIGURE 8. **Facial landmark localisation**

5) STEP 5- SELECTING EYE LANDMARKS FROM THE 68 LANDMARKS.

The eyes and the eye area are the most critical parts of detecting drowsiness. Most drowsiness detection systems detect drowsiness by comparing eye conditions and eyelid movements. The facts have shown that the eyes are an essential factor in determining a person's degree of attention to his surroundings. When a person is tired from a lack of rest or physical effort, they lose concentration. In this case, if the same person drives the vehicle, they are prone to accidents. There are six coordinates in Figure 9 around each eye.

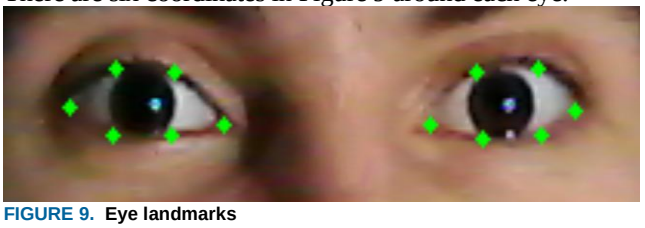


FIGURE 9. **Eye landmarks**

Those coordinates indicate each eye, starting at the left corner and working clockwise around the rest of the eye. These 12 points (6 of each eye) are joined to make a circumference around the eye (Figure 10) and then used to calculate ECR for drowsiness detection.

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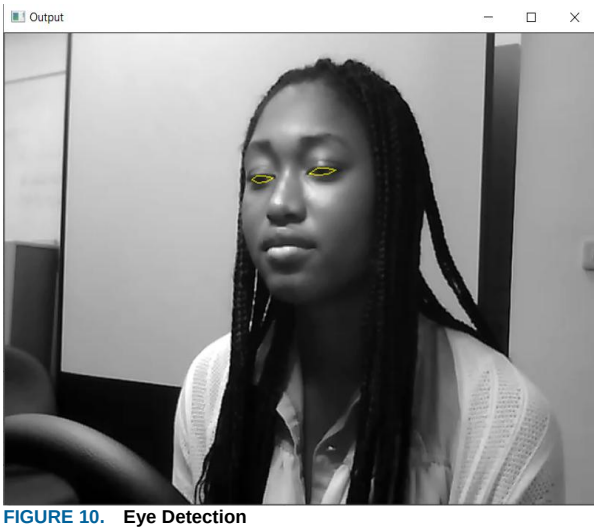


FIGURE 10. **Eye Detection**

6) STEP 6- CALCULATING ADAPTIVE THRESHOLD

The next step after identifying the eyes is to detect drowsiness in the retrieved frames. We live in a world with an estimated population of 8 billion people that is rich in diversity. The same variety can be seen in the eyes as well. Since there are so many different eye shapes and forms globally, finding a standard threshold point for all of them becomes almost impossible. The solution to this issue is adaptive thresholding, which ensures that the threshold value for each person will be different since it will be extracted from that individual's eye shape. The postulated adaptive thresholding approach, as well as its computations on the equations (14-20) in step 7 of the same section, adds uniqueness to our study. A novel idea of computing eye aspect ratio for driver drowsiness detection also adds stability to the system. The videos of the NTHU-DDD Dataset are fed to the system. The driver is instructed to close his eyes for 15 seconds and then open his eyes for another 15 seconds. This is called the setup phase. In the setup phase, out of the 900 frames, as processed in data pre-processing, the mean of the top 50 frames with the greatest eye characteristic ratio (no eye-closing instances) and the bottom 50 frames, i.e., with the lowest eye characteristic ratio (no eye-opening instances) is computed. The average of both calculated means is set as the subject's threshold value (Figure 11). Following the evaluation of the threshold value, the model determines whether the driver is sleepy or not based on the comparison of threshold and ECR. Figure 12 represents the disparity in the threshold values which occur due to the different shapes and sizes of eyes of the subjects taken. So, it is not suitable to set a constant threshold value as it will provide undesirable results.

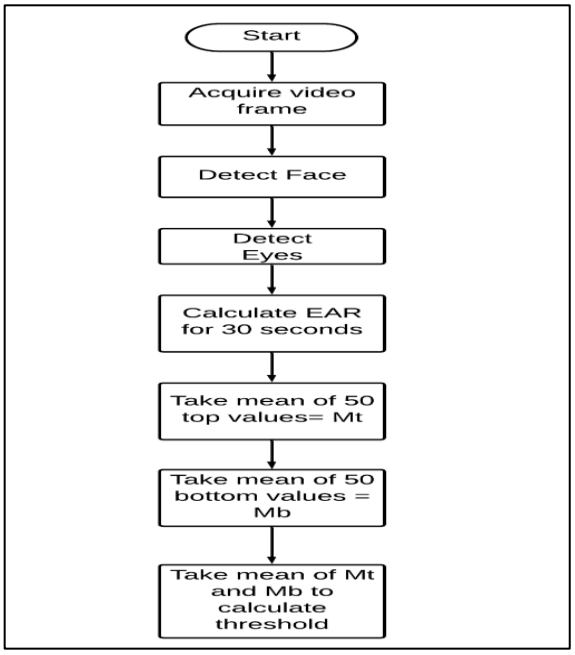


FIGURE 11. **Flowchart of adaptive thresholding**

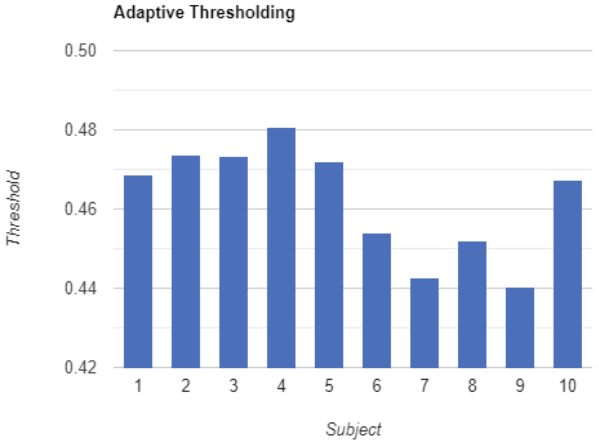


FIGURE 12. **Variations in the threshold of the subjects**

7) STEP 7- CALCULATING ECR

To identify drowsiness, the EAR is often defined as the ratio of the height and width of the eye. We have used the existing formulas of EFR [30][33] and named it the Eye Characteristic Ratio (ECR), which was used to detect blinking. We tested it to calculate the eye-opening/closing ratio of the eye for drowsiness detection.

The numerator represents the eye's diagonal height, while the denominator represents the eye's width. Figures 13a and 13b show the specifics of all the landmarks of the open and closed eyes, respectively. As per our study, no prior research takes the diagonal distance between landmarks to determine a person's drowsiness.

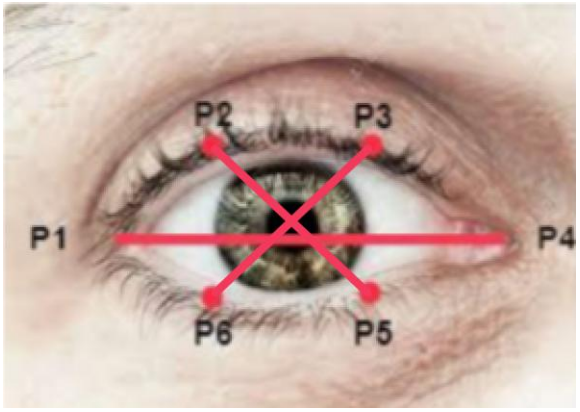
The numerator value rises when the eyes are open, raising the ECR value, but the numerator value decreases when the eyes are closed, reducing the ECR value. ECR is calculated for each frame of the video stream. The average of the ECR values of the left and right eye is determined. The ECR is essentially constant for an open eye, while it is near to zero for a closed eye. The ECR of the pair is average since both blinking and opening/closing the eyes are done by both eyes.

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(B)



(A)

FIGURE 13. **A: Open Eye with landmarks detected and B: Closed Eye with landmarks detected**

$$ECR_{L1}=|p2_L - p5_L| \quad (14)$$

$$ECR_{L2}=|p3_L - p6_L| \quad (15)$$

$$ECR_L = \frac{|ECR_{L1} + ECR_{L2}|}{2|P1_L - P4_L|} \quad (16)$$

$$ECR_{R1}=|p2_R - p5_R| \quad (17)$$

$$ECR_{R2}=|p3_R - p6_R| \quad (18)$$

$$ECR_R = \frac{|ECR_{R1}| + ECR_{R2}|}{2|P1_R - P4_R|} \quad (19)$$

$$ECR_F = \frac{|ECR_L + ECR_R|}{2} \quad (20)$$

Figure 14 represents the ECR of each frame for the entire video upon implementing the formulas as mentioned above.

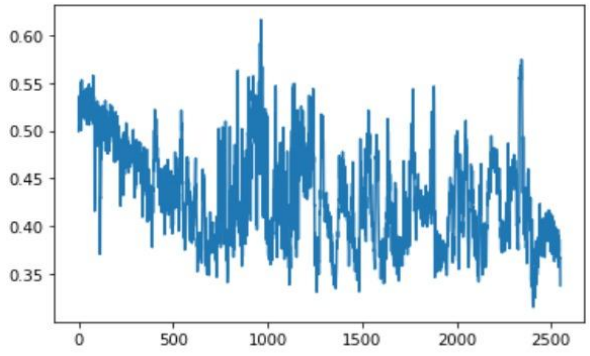
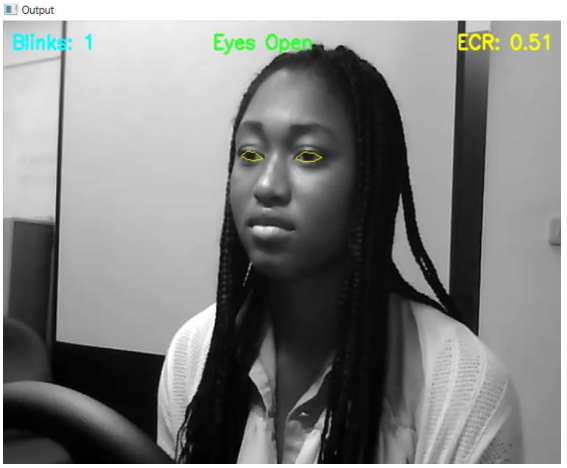
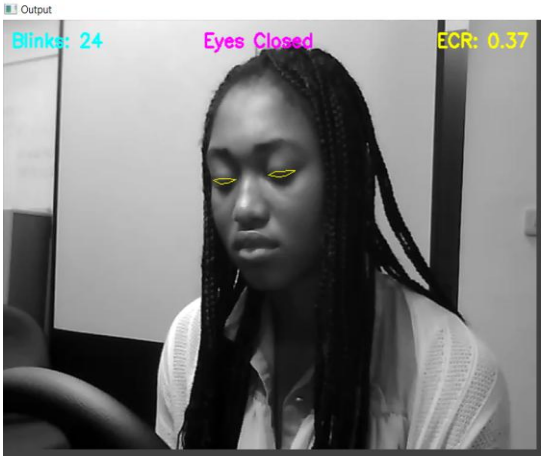


FIGURE 14. **ECR of a Subject**

As illustrated in Figure 15a, the value of open eyes ECR will be higher than the threshold. On the other hand, the value for closed eyes will be less than the threshold, as shown in Figure 15b.



(A)



(B)

FIGURE 15. **A: Open Eye ECR Evaluation and B: Closed Eye ECR Evaluation**

8) STEP 8- DETECTION OF BLINK DEPENDING UPON THE NUMBER OF FRAMES.

Everyone exhibits a different blink pattern which differs in speed, duration, and the degree to which the eye is squeezed. The eye blink lasts for approximately 100-400 milliseconds [30][32]. Frames from a video series are extracted and subsequently processed to recognize the face and hence the eye component of the image to calculate the blink rate per minute. In our system, blinks are determined based on the number of frames. If the eyelids remain closed for less than 70 frames for a 30-fps video, blinks are recognized, and the output indicates the number of blinks. If the total count of blinks is less than 10 per minute, then the system identifies tiredness, as illustrated in Figure 16.

9) STEP 9- DETECTION OF DROWSINESS DEPENDING UPON THE NUMBER OF FRAMES HENCE ALERTING THE DRIVER

The algorithm's final step is to use the ECR value to assess the driver's condition. An individual's blink lasts 100 to 400

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milliseconds on average [30]. This takes between 0.1 and 0.4 seconds. As a result, if a person feels drowsy, their eyelids should be closed for a longer period than this. If for 70 frames or more the driver's eyes are closed, an alarm will sound to warn the driver [31]. Figure 17 shows a drowsiness warning.

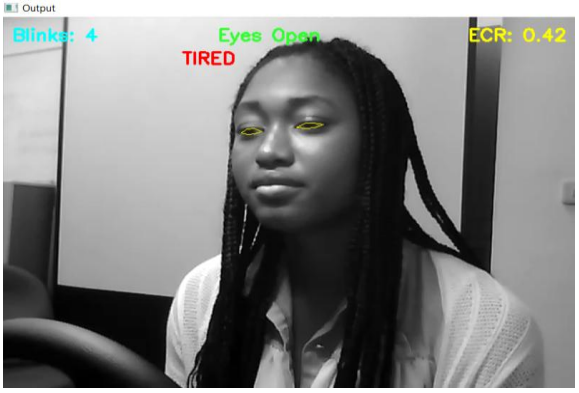


FIGURE 16. **Tiredness Warning**



FIGURE 17. **Drowsiness Alert**

IV. Results

Section 1 briefly explained the significance of the need for an effective driver drowsiness detection system. Section 2 consists of the associated work-study. Section 3 provides a step-by-step exposition of the entire model, the calculations, and evaluations for adaptive thresholding, eye blinking, and drowsiness detection. The suggested system was created and tested using the dataset. Following that, the feature values are saved for statistical analysis and categorization.

A. Result Description

Four classification algorithms (Random Forest, Decision Tree, KNN, and Multinomial) were used to classify the data. Sensitivity, specificity, precision, accuracy, negative predictive value, false-positive rate, false discovery rate, false-negative rate, F1 score, Recall, MSE, RMSE, and Matthews Correlation Coefficient are evaluated. Additionally, ROC plots are created, and AUC is calculated to analyse the proposed system's performance. Detailed averaged comparative analysis depicting Sensitivity, Precision, Specificity, Negative Predictive Value, False Positive Rate, False Negative Rate, Accuracy, F1 score, Recall, MSE, RMSE, and Matthew's correlation coefficient are shown in Table 4.

B. Result Analysis

The suggested system for detecting driver drowsiness and eye blinking is developed using specific OpenCV methods, and then several algorithms were evaluated, yielding the findings shown below. The presented result relies on each frame of the video for blink as computed using the formulae in (Sec 3). The Confusion matrix of blinking accuracy is presented for the testing stage. The accuracy for blink detection is 99.784%.

Figures 18, 19, 20, 21 show the ROC Curve for Random Forest, Decision Tree, KNN, and Multinomial classifiers, respectively. Below mentioned are their respective AUC values. The Random Forest Classifier achieved the highest value of AUC (0.9088). Table 4 gives the detailed comparative analysis of the four classifiers (Random Forest, Decision Tree, KNN, and Multinomial classifier) used on the proposed system. The random forest classifier provides the highest accuracy when drowsiness is identified, as shown in Table 4.

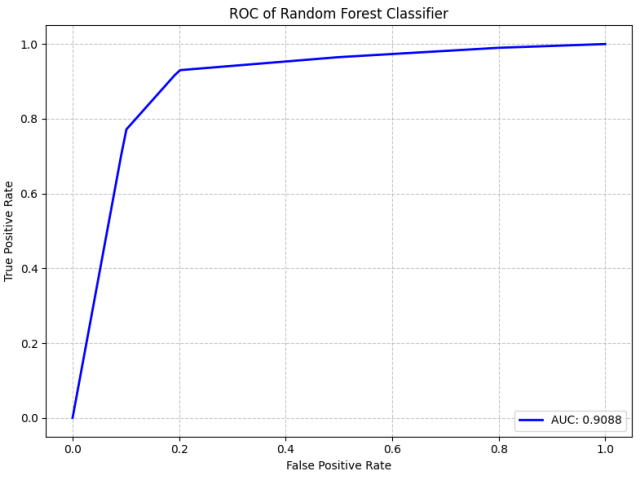


FIGURE 18. **ROC of Random Forest Classifier**

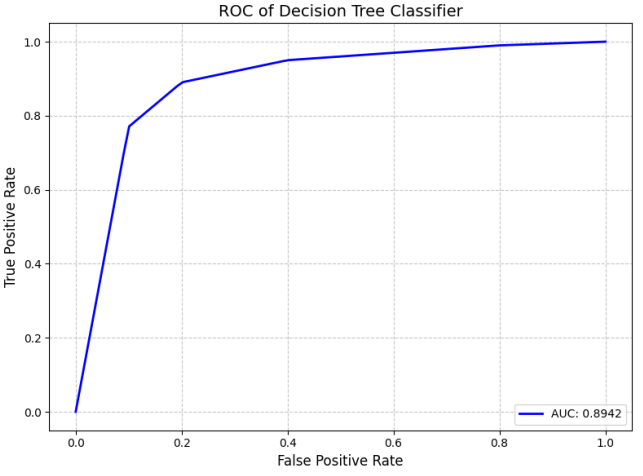


FIGURE 19. **ROC of Decision Tree Classifier**

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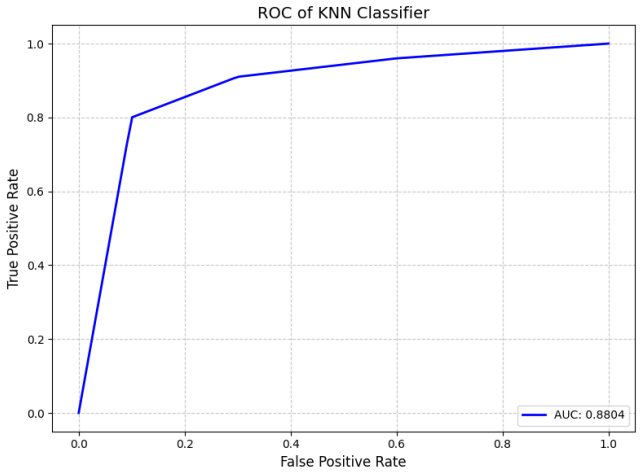


FIGURE 20. ROC of KNN Classifier

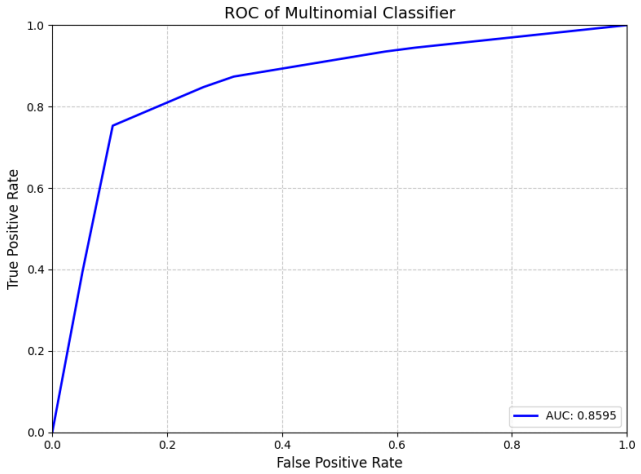


FIGURE 21. ROC of Multinomial Classifier

TABLE 4
AVERAGED COMPARATIVE ANALYSIS

Measures	Random Forest	Decision Tree	KNN	Multinomial
Sensitivity	0.8711	0.8440	0.8491	0.8677
Specificity	0.9653	0.9800	0.9444	0.8168
Precision	0.9746	0.9863	0.9596	0.8367
Negative Predictive Value	0.8305	0.7871	0.8006	0.8509
False Positive Rate	0.0347	0.0200	0.0556	0.1832
False Discovery Rate	0.0254	0.0137	0.0404	0.1633
False Negative Rate	0.1289	0.1560	0.1509	0.1323
Accuracy	0.9048	0.8944	0.8863	0.8432
F1 Score	0.9200	0.9096	0.9010	0.8519
Recall	0.8387	0.7999	0.7929	0.8734
Loss	0.0971	0.1072	0.1146	0.1642
RMSE	0.3116	0.3274	0.3385	0.4052
MCC	0.8207	0.7983	0.7767	0.6860

1. Multinomial Naive Bayes is a potent that is being used to analyse text input and solve issues involving numerous classes. It's based on the formula below:

$$P(A|B) = P(A) * P(B|A)/P(B) \quad (21)$$

For the calculation of the probability of class A when predictor B is already provided.
P(B) = prior probability of B
P(A) = prior probability of class A
P(B|A) = occurrence of predictor B
This formula aids in determining the probability of tags in the text.

2. The Random Forest Classifier is constituted of several distinct decision trees, all of which have the same nodes but use different data to produce diverse leaves. It combines the results of numerous decision trees to arrive at a single solution that represents the average of all of them.

$$MSE=\frac{1}{N}\sum_{i=1}^{N}(f_i - y_i)^2 \quad (22)$$

where **N** is the total number of data points, f_i is the actual value of the ith data point, and y_i is the predicted value of the ith data point.

$$Gini=1-\sum_{i=1}^{N}(p_i)^2 \quad (23)$$

$$Entropy=\sum_{i=1}^{N} -p_i * log_2(p_i) \quad (24)$$

where p_i is probability of class i occurring in the dataset and N refers to the number of possible categories.

3. A decision tree constructs classification or regression models in the similar manner of a tree structure. It gradually cuts down a dataset into smaller and smaller sections while also developing an associated decision tree simultaneously. With decision nodes and leaf nodes, this is the finished vision-based tree. Information Entropy for a dataset with C classes is:

$$Entropy=\sum_{i=1}^{C} -p_i log_2 p_i \quad (25)$$

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where $\boldsymbol{p_i}$ is the probability of randomly picking an element of class *i*.

$$E(T,X)=\sum_{c\in X} \boldsymbol{P(c)}\,\boldsymbol{E(c)} \quad (26)$$

Equation 26 calculates the **weighted entropy** after splitting a dataset T based on a feature X where
c∈X: Each subset after the split,
P(c): Proportion of instances in subset c,
E(c): Entropy of subset ccc.

$$Gini(T,X)=Entropy(T)-Entropy(T,X) \quad (27)$$

Equation 27 represents the **reduction in entropy** achieved by splitting T into subsets X where
Entropy(T): The entropy of the entire dataset before the split.
Entropy(T,X): The weighted entropy of the subsets after the split.

4. KNN refers to K-nearest neighbour and is a supervised learning method that is commonly used to classify data based on how its neighbours are classified. KNN keeps track of all available instances and categorises new ones using a similarity metric. The parameter K in KNN denotes the number of closest neighbours to consider in the majority voting procedure.

Manhattan

$$d(x,y)=\sum_{i=1}^{n} |\boldsymbol{x_i}-\boldsymbol{y_i}| \quad (28)$$

Euclidean

$$d(x,y)=\sqrt{\sum_{i=1}^{n}(\boldsymbol{x_i}-\boldsymbol{y_i})^2} \quad (29)$$

Minkowski

$$d(x,y)=\left(\sum_{i=1}^{n}|\boldsymbol{x_i}-\boldsymbol{y_i}|^{c}\right)^{\frac{1}{c}} \quad (30)$$

The Random Forest classifier achieved the best accuracy of 90.48 per cent, with the lowest loss per cent among the four classifiers. Random forest achieved the greatest accuracy of all four classifiers due to its tremendous benefits.

- Random forests are considered a particularly accurate and robust technique due to the vast number of decision trees present in them.
- It is unaffected by the overfitting problem. The primary reason for this is that it takes the average of all forecasts, balancing any biases.
- Random forests can also manage missing values.

Figure 22 shows a comparative study of the ROC for each classifier.

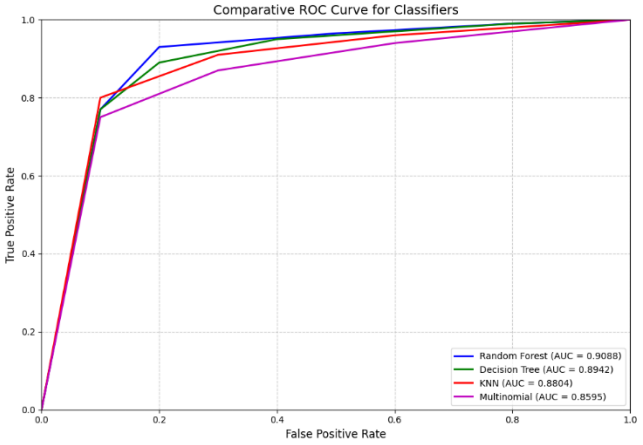


FIGURE 22. Comparative ROC

V. Conclusion

This research is mainly concerned with detecting driver drowsiness via vision. This paper suggests a system to monitor and identify driver distraction, recognize the driver's face by collecting facial landmarks and warning the driver to avoid crashes. The proposed technique detects driver drowsiness by combining adaptive threshold, eye characteristic ratio, and eye blink count. It presents a combination of two levels of the state of the driver while driving.

- According to the first level, the eye blink detection and blink count identify drowsiness, which is critical in determining whether a person is fatigued or not. As a result, driving should be avoided in such circumstances to avoid accidents.
- The next stage involves detecting the driver's drowsiness based on a threshold and ECR value comparison in a time frame of 2 seconds and alerting the driver with an alarm if drowsiness is found. The suggested system makes use of data acquired under various situations.

Different machine learning classifiers are applied to the constructed system. Thus, a comparison of accuracy, loss, and certain other parameters when working with different classifiers is provided, displaying the best-suited classifier for our model, bringing the work to an end.

A. Future Work

The results are satisfactory, but there is potential for improvement in face identification by including new methods and characteristics to construct a more robust and more efficient model. Several additional innovative techniques and enhanced learning may also be added to achieve the most satisfactory results. Future research may use other facial characteristics for drowsiness evaluation, such as the mouth for yawning and the head position estimation for nodding. Currently, there is no way to change the camera's zoom or orientation while it is in use. It's also possible that future work may include automatically zooming in on the face after it's been located. Furthermore, we will attempt to build our algorithm on a DSP (Digital Signal Processor) to construct a self-contained system that operates in real-time. It is expected

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that once the developers test the gadget in a car, any design problems will be found, and additional data on the device's actual operation in real scenarios will be obtained.

Conflict of interest statement
No conflicts of interest.

Data Availability Statement
Data is available at http://cv.cs.nthu.edu.tw/php/callforpaper/datasets/DDD/

Ethical statement
No research on human subjects was involved.

Funding statement
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